Topics

Authentication

Authentication is the process of verifying that a user is who they claim to be before granting access to an account or system. It helps protect personal information and digital services by checking credentials such as passwords, security keys, or biometric data. (**Matteo)**

- **Authentication Fundamentals (Matteo)**
  How authentication works, how it differs from identification and authorisation, and the three main authentication factors: something you know, something you have, and something you are.
- **Passwords and Passphrases (Alejandro)**
  How password length, complexity, and reuse affect account security, and how to create stronger, memorable passphrases.
- **Adaptive Authentication (Giulio)**
  How authentication systems adapt their checks based on risk signals such as the device, location, and login behaviour, balancing account security with user convenience.
- **Biometric Authentication (Giulio)**
  How fingerprints, facial recognition, and other biological characteristics verify identity, with a focus on reliability and privacy concerns.
- **Passwordless Authentication and Passkeys (Giulio)**
  How passkeys allow users to sign in without traditional passwords and help protect against phishing and stolen credentials.
- **Password Managers (Yho)**
  How password managers generate, store, and autofill credentials, including their benefits, potential risks, and safe usage.
- **Phishing and Fake Login Pages (Yho)**
  How attackers trick users into revealing login credentials, how to recognise suspicious login requests, and how websites can protect users.
- **Attacks Against Password-Based Authentication (Atakan)**
  How brute-force attacks, dictionary attacks, and credential stuffing work, and how rate limits and other safeguards reduce their effectiveness.
- **Single Sign-On and Social Login (Atakan)**
  How signing in through one identity provider simplifies access to multiple services, and what this means for security and personal data sharing.
- **Account Recovery and Password Resets (Atakan)**
  How users regain access to their accounts, why recovery procedures can become security weaknesses, and how to design safer alternatives.
- **Sessions, Cookies, and Staying Logged In (Atakan)**
  How websites remember authenticated users, how session hijacking can compromise accounts, and why secure cookies, session expiration, and logout matter.

Topics

Risk broked by AI

Artificial intelligence can introduce or amplify risks to cybersecurity and digital privacy. These risks include the malicious use of AI, the exposure of personal information, and weaknesses within AI systems themselves. This section explores these threats and ways to reduce their impact. **(Matteo)**

- **AI-Generated Phishing and Social Engineering (Matteo)**
  How AI can produce convincing, personalised scams at scale, making deceptive emails and messages harder to recognise.
- **Deepfakes and Identity Impersonation (Sezer)**
  How synthetic voices and videos can impersonate real people, enable fraud, and undermine identity verification.
- **AI-Assisted Malware and Cyberattacks (Yho)**
  How attackers can use AI to assist with writing malicious code and automating parts of cyberattacks, including the limitations of these tools.
- **Sensitive Data Shared with AI Chatbots (Yho)**
  How entering personal, academic, or confidential information into AI services can expose sensitive data, and how users can reduce these risks.
- **AI-Powered Surveillance and Facial Recognition (Giulio)**
  How AI enables large-scale monitoring and identification, raising concerns about consent, anonymity, and the misuse of personal information.
- **Personal Data Profiling and Inference (Alejandro)**
  How AI can combine seemingly harmless data to infer sensitive information about people, such as their habits, interests, or health.
- **AI-Generated Disinformation (Alejandro)**
  How AI-generated text, images, and videos can spread false information, manipulate audiences, and weaken trust in digital content.
- **Bias and Discrimination in AI Security Systems (Alejandro)**
  How biased AI systems can unfairly flag users as suspicious or deny legitimate access, and why testing and human oversight matter.
- **Prompt Injection and AI System Manipulation (Sezer)**
  How malicious instructions hidden in user inputs or external content can redirect AI systems, potentially causing information leaks or unauthorised actions.
- **Data Poisoning and AI Model Backdoors(Sezer)**
  How tampering with training data can influence an AI system's behaviour, reduce its reliability, or introduce hidden vulnerabilities.
- **AI Hallucinations and Overreliance in Cybersecurity (Sezer)**
  How incorrect AI-generated advice or analysis can lead to unsafe decisions, and why security recommendations require independent verification.